

ANMOL KUMAR

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EDUCATION

Master of Computer Application

2021-2024

National Institute Of Technology, Jamshedpur

EXPERIENCE

CyborgIntell Pvt. Ltd.

Jan 2024 — Present

Software Engineer (AI/ML)

HSR Layout, Bengaluru

- Architected and developed a comprehensive **ML Model Inference Service** in Java leveraging **Deep Java Library (DJL)**, supporting model upload, loading, and multi-modal inference capabilities including batch processing, real-time predictions, and scheduled periodic inference with integrated job scheduling.
- Built a complete **Accounts Receivable Solution** in Java featuring prediction analytics, automated action recommendations, and collection management modules, consuming ML inference outputs to streamline financial workflows and debt recovery processes.
- Designed and implemented a **Model Performance Monitoring Service** with comprehensive metrics tracking for diverse ML model types, enabling real-time performance analysis, drift detection, and automated alerting for model degradation.
- Created a **Retrieval-Augmented Generation (RAG) System** for intelligent query analysis across diverse document types (PDF, DOCX, TXT, etc.), integrating vector databases and LLM-based inference to deliver accurate, context-aware responses and knowledge retrieval.
- Orchestrated containerized applications using **Kubernetes**, implementing automated scaling, service discovery, and rolling deployments across development and production environments for all developed microservices.
- Designed scalable REST APIs using **FastAPI** and **Jersey Java** framework, enabling seamless integration between ML inference services, account management systems, and container management tools.
- Optimized deployment pipelines reducing application deployment time by 20% through advanced **Docker** containerization, multi-stage builds, and efficient registry management strategies.
- Enhanced overall system performance by 15% through strategic **Nginx** configuration, reverse proxy optimization, and intelligent load balancing across Kubernetes clusters.

PROJECTS

RAG Assistant (Basic Retrieval-Augmented Generation App)

- Built a **Retrieval-Augmented Generation (RAG)** pipeline using **LangChain**, **LanceDB**, and **Ollama API**.
- Implemented document ingestion, embedding generation, and context-aware response generation.
- Deployed as a lightweight service for interactive Q&A over custom documents.

Docker Engine API Client (Java)

- Open-source contribution: Built a **Java** client library for Docker Engine API, framework-agnostic.
- Supports container lifecycle management operations: create, start, stop, remove.
- Designed for integration into any Java-based project for container orchestration tasks.

LanceDB Java Client

- Developed a **Java** client SDK for **LanceDB**, enabling vector database integration.
- Implemented APIs for vector upserts, similarity search, and collection management.
- Built with focus on performance and easy integration into enterprise Java applications.

TECHNICAL SKILLS

- **Programming Languages:** Java , GoLang, Python
- **Backend Frameworks:** Jersey, Gin , FastApi
- **Cloud Containerization:** Kubernetes , Docker , Nginx
- **Databases:** MySQL, SQL optimization, schema design
- **Development Tools:** Git, IntelliJ IDEA, Eclipse, Postman
- **Core Subjects:** Data Structures Algorithms, Object-Oriented Programming, Operating Systems, DBMS